

**Topics:** 13.2 Derivatives and Integrals of Vector Functions; 13.3 Arc Length and Curvature; 13.4 Motion in Space: Velocity and Acceleration

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- Find the derivative of the vector function

$$\mathbf{r}(t) = \left\langle e^{t^2+2t}, \ln(\cos(t)), t \arctan(t) \right\rangle.$$

We differentiate each component:

$$\frac{d}{dt} \left( e^{t^2+2t} \right) = (2t+2)e^{t^2+2t},$$

$$\frac{d}{dt} (\ln(\cos t)) = \frac{-\sin t}{\cos t} = -\tan t,$$

$$\frac{d}{dt} (t \arctan t) = \arctan t + \frac{t}{t^2+1}.$$

Therefore,

$$\mathbf{r}'(t) = \left\langle (2t+2)e^{t^2+2t}, -\tan(t), \arctan(t) + \frac{t}{t^2+1} \right\rangle.$$

- Evaluate the definite integral

$$\int_0^1 \left( \frac{1}{t+1} \mathbf{i} + \frac{1}{t^2+1} \mathbf{j} + \frac{t}{t^2+1} \mathbf{k} \right) dt.$$

We integrate each component:

$$\int_0^1 \frac{1}{t+1} dt = [\ln|t+1|]_0^1 = \ln(2) - \ln(1) = \ln(2),$$

$$\int_0^1 \frac{1}{t^2+1} dt = [\arctan(t)]_0^1 = \arctan(1) - \arctan(0) = \frac{\pi}{4},$$

$$\int_0^1 \frac{t}{t^2+1} dt = \frac{1}{2} \int_0^1 \frac{2t}{t^2+1} dt = \frac{1}{2} [\ln(t^2+1)]_0^1 = \frac{1}{2} (\ln(2) - \ln(1)) = \frac{1}{2} \ln(2).$$

Therefore,

$$\int_0^1 \left( \frac{1}{t+1} \mathbf{i} + \frac{1}{t^2+1} \mathbf{j} + \frac{t}{t^2+1} \mathbf{k} \right) dt = \ln(2) \mathbf{i} + \frac{\pi}{4} \mathbf{j} + \frac{1}{2} \ln(2) \mathbf{k}.$$

- Find a vector equation of the tangent line to the curve

$$\mathbf{r}(t) = t \cos(2t) \mathbf{i} + 4t \mathbf{j} + \sin(t) \mathbf{k}$$

at the point corresponding to  $t = \frac{\pi}{4}$ .

First, evaluate  $\mathbf{r}(t)$  at  $t = \frac{\pi}{4}$ :

$$\mathbf{r}\left(\frac{\pi}{4}\right) = \frac{\pi}{4} \cos\left(2 \cdot \frac{\pi}{4}\right) \mathbf{i} + 4 \cdot \frac{\pi}{4} \mathbf{j} + \sin\left(\frac{\pi}{4}\right) \mathbf{k} = \pi \mathbf{j} + \frac{\sqrt{2}}{2} \mathbf{k}.$$

Next, differentiate  $\mathbf{r}(t)$ :

$$\mathbf{r}'(t) = (\cos(2t) - 2t \sin(2t)) \mathbf{i} + 4 \mathbf{j} + \cos(t) \mathbf{k}.$$

Evaluating at  $t = \frac{\pi}{4}$  gives

$$\mathbf{r}'\left(\frac{\pi}{4}\right) = \left(\cos\left(\frac{\pi}{2}\right) - 2 \cdot \frac{\pi}{4} \sin\left(\frac{\pi}{2}\right)\right) \mathbf{i} + 4 \mathbf{j} + \cos\left(\frac{\pi}{4}\right) \mathbf{k} = -\frac{\pi}{2} \mathbf{i} + 4 \mathbf{j} + \frac{\sqrt{2}}{2} \mathbf{k}.$$

Therefore, a vector equation for the tangent line is

$$\mathbf{L}(t) = \left(\pi \mathbf{j} + \frac{\sqrt{2}}{2} \mathbf{k}\right) + t \left(-\frac{\pi}{2} \mathbf{i} + 4 \mathbf{j} + \frac{\sqrt{2}}{2} \mathbf{k}\right).$$

4. Find the length of the curve  $\mathbf{r}(t) = 2t \mathbf{i} + t^2 \mathbf{j} + \frac{1}{3}t^3 \mathbf{k}$  over the interval  $0 \leq t \leq 1$ .

First compute the derivative:

$$\mathbf{r}'(t) = 2 \mathbf{i} + 2t \mathbf{j} + t^2 \mathbf{k}.$$

Now compute its magnitude:

$$\begin{aligned} |\mathbf{r}'(t)| &= \sqrt{(2)^2 + (2t)^2 + (t^2)^2} \\ &= \sqrt{4 + 4t^2 + t^4} \\ &= \sqrt{(2 + t^2)^2} \\ &= 2 + t^2. \end{aligned}$$

Using the arc length formula  $L = \int_0^1 |\mathbf{r}'(t)| dt$ , we get

$$\begin{aligned} L &= \int_0^1 (2 + t^2) dt \\ &= \left[2t + \frac{1}{3}t^3\right]_0^1 \\ &= \left(2 + \frac{1}{3}\right) - 0 \\ &= \frac{7}{3}. \end{aligned}$$

5. A particle starts at the point  $(0, 0, 3)$  at time  $t = 0$  and moves with position function

$$\mathbf{r}(t) = \langle 3 \sin t, 4t, 3 \cos t \rangle.$$

Find the position of the particle after it has traveled a distance of  $5\pi$  units.

To find the time  $T$  when the particle has traveled  $5\pi$  units, we solve

$$5\pi = \int_0^T |\mathbf{r}'(t)| dt.$$

Differentiate  $\mathbf{r}(t)$ :

$$\mathbf{r}'(t) = \langle 3 \cos t, 4, -3 \sin t \rangle.$$

Now compute the speed:

$$\begin{aligned} |\mathbf{r}'(t)| &= \sqrt{(3 \cos t)^2 + 4^2 + (-3 \sin t)^2} \\ &= \sqrt{9 \cos^2 t + 16 + 9 \sin^2 t} \\ &= \sqrt{9(\sin^2 t + \cos^2 t) + 16} \\ &= \sqrt{9 + 16} \\ &= 5. \end{aligned}$$

So the equation we are solving becomes

$$5\pi = \int_0^T 5 dt = 5T,$$

which gives  $T = \pi$ . The position at that time is

$$\mathbf{r}(\pi) = \langle 3 \sin \pi, 4\pi, 3 \cos \pi \rangle = \langle 0, 4\pi, -3 \rangle.$$

6. Consider the vector function  $\mathbf{r}(t) = \langle t, t^2, 4 \rangle$ .

(a) Find the unit tangent vector  $\mathbf{T}(t)$ .

We begin by computing  $\mathbf{r}'(t)$  and its magnitude:

$$\mathbf{r}'(t) = \langle 1, 2t, 0 \rangle, \quad |\mathbf{r}'(t)| = \sqrt{1 + (2t)^2} = \sqrt{1 + 4t^2}.$$

The unit tangent vector is

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} = \frac{\langle 1, 2t, 0 \rangle}{\sqrt{1 + 4t^2}}.$$

(b) Find the unit normal vector  $\mathbf{N}(t)$ .

The unit normal vector is defined by

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{|\mathbf{T}'(t)|}.$$

Differentiate  $\mathbf{T}(t)$ :

$$\begin{aligned}\mathbf{T}'(t) &= \frac{1}{\sqrt{1+4t^2}} \langle 0, 2, 0 \rangle - \frac{1}{2}(1+4t^2)^{-3/2} \cdot 8t \langle 1, 2t, 0 \rangle \\ &= \frac{1}{(1+4t^2)^{3/2}} \left[ (1+4t^2) \langle 0, 2, 0 \rangle - 4t \langle 1, 2t, 0 \rangle \right] \\ &= \frac{\langle -4t, 2, 0 \rangle}{(1+4t^2)^{3/2}}.\end{aligned}$$

Now compute its magnitude:

$$\begin{aligned}|\mathbf{T}'(t)| &= \frac{\sqrt{(-4t)^2 + 2^2}}{(1+4t^2)^{3/2}} = \frac{\sqrt{16t^2 + 4}}{(1+4t^2)^{3/2}} \\ &= \frac{2\sqrt{1+4t^2}}{(1+4t^2)^{3/2}} = \frac{2}{1+4t^2}.\end{aligned}$$

Therefore,

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{|\mathbf{T}'(t)|} = \frac{\langle -4t, 2, 0 \rangle}{(1+4t^2)^{3/2}} \cdot \frac{1+4t^2}{2} = \frac{\langle -2t, 1, 0 \rangle}{\sqrt{1+4t^2}}.$$

(c) Find the curvature  $\kappa(t)$ .

The curvature is

$$\kappa(t) = \frac{|\mathbf{T}'(t)|}{|\mathbf{r}'(t)|} = \frac{2/(1+4t^2)}{\sqrt{1+4t^2}} = \frac{2}{(1+4t^2)^{3/2}}.$$

7. Find the vectors  $\mathbf{T}$ ,  $\mathbf{N}$ , and  $\mathbf{B}$  at the point  $(1, \frac{2}{3}, 1)$  for the vector function

$$\mathbf{r}(t) = \langle t^2, \frac{2}{3}t^3, t \rangle.$$

Since  $\mathbf{r}(t) = \langle t^2, \frac{2}{3}t^3, t \rangle$ , the point  $(1, \frac{2}{3}, 1)$  corresponds to  $t = 1$ . First compute

$$\mathbf{r}'(t) = \langle 2t, 2t^2, 1 \rangle.$$

Then the unit tangent vector is

$$\begin{aligned}\mathbf{T}(t) &= \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} = \frac{\langle 2t, 2t^2, 1 \rangle}{\sqrt{(2t)^2 + (2t^2)^2 + 1}} \\ &= \frac{\langle 2t, 2t^2, 1 \rangle}{\sqrt{4t^2 + 4t^4 + 1}} = \frac{\langle 2t, 2t^2, 1 \rangle}{2t^2 + 1}.\end{aligned}$$

So,

$$\mathbf{T}(1) = \left\langle \frac{2}{3}, \frac{2}{3}, \frac{1}{3} \right\rangle.$$

Next, differentiate  $\mathbf{T}(t)$ :

$$\begin{aligned}\mathbf{T}'(t) &= \frac{\langle 2, 4t, 0 \rangle}{2t^2 + 1} - (2t^2 + 1)^{-2} \cdot 4t \langle 2t, 2t^2, 1 \rangle \\ &= \frac{1}{(2t^2 + 1)^2} \left[ \langle 4t^2 + 2, 8t^3 + 4t, 0 \rangle - \langle 8t^2, 8t^3, 4t \rangle \right] \\ &= \frac{2}{(2t^2 + 1)^2} \langle 1 - 2t^2, 2t, -2t \rangle.\end{aligned}$$

Thus,

$$\begin{aligned}\mathbf{N}(t) &= \frac{\mathbf{T}'(t)}{|\mathbf{T}'(t)|} = \frac{\langle 1 - 2t^2, 2t, -2t \rangle}{\sqrt{(1 - 2t^2)^2 + (2t)^2 + (-2t)^2}} \\ &= \frac{\langle 1 - 2t^2, 2t, -2t \rangle}{\sqrt{1 + 4t^2 + 4t^4}} = \frac{\langle 1 - 2t^2, 2t, -2t \rangle}{1 + 2t^2}.\end{aligned}$$

At  $t = 1$ ,

$$\mathbf{N}(1) = \left\langle -\frac{1}{3}, \frac{2}{3}, -\frac{2}{3} \right\rangle.$$

Finally, the binormal vector is  $\mathbf{B} = \mathbf{T} \times \mathbf{N}$ , so

$$\mathbf{B}(1) = \mathbf{T}(1) \times \mathbf{N}(1) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{2}{3} & \frac{2}{3} & \frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} & -\frac{2}{3} \end{vmatrix} = \left\langle -\frac{2}{3}, \frac{1}{3}, \frac{2}{3} \right\rangle.$$

8. A particle moves in space with position function

$$\mathbf{r}(t) = \langle t \ln t, t, e^{-t} \rangle$$

for  $t > 0$ . Find the velocity, speed, and acceleration of the particle.

The velocity is  $\mathbf{v}(t) = \mathbf{r}'(t)$ , so

$$\mathbf{v}(t) = \langle \ln(t) + 1, 1, -e^{-t} \rangle.$$

The acceleration is  $\mathbf{a}(t) = \mathbf{v}'(t)$ , so

$$\mathbf{a}(t) = \left\langle \frac{1}{t}, 0, e^{-t} \right\rangle.$$

Finally, the speed is the magnitude of the velocity:

$$\|\mathbf{v}(t)\| = \sqrt{(\ln(t) + 1)^2 + 1^2 + (-e^{-t})^2} = \sqrt{(\ln(t) + 1)^2 + 1 + e^{-2t}}.$$

9. The acceleration of an object is given by  $\mathbf{a}(t) = t\mathbf{i} + e^t\mathbf{j} + e^{-t}\mathbf{k}$ . Find the object's position function  $\mathbf{r}(t)$  if the initial velocity is  $\mathbf{v}(0) = \mathbf{k}$  and the initial position is  $\mathbf{r}(0) = \mathbf{j} + \mathbf{k}$ .

Since  $\mathbf{a}(t) = \mathbf{v}'(t)$  and  $\mathbf{v}(t) = \mathbf{r}'(t)$ , we first find  $\mathbf{v}(t)$  by integrating  $\mathbf{a}(t)$ :

$$\begin{aligned}\mathbf{v}(t) &= \int \mathbf{a}(t) dt = \int (t\mathbf{i} + e^t\mathbf{j} + e^{-t}\mathbf{k}) dt \\ &= \frac{t^2}{2}\mathbf{i} + e^t\mathbf{j} - e^{-t}\mathbf{k} + \mathbf{C}_1.\end{aligned}$$

Use  $\mathbf{v}(0) = \mathbf{k}$ :

$$\mathbf{v}(0) = \langle 0, 1, -1 \rangle + \mathbf{C}_1 = \langle 0, 0, 1 \rangle \Rightarrow \mathbf{C}_1 = 2\mathbf{k} - \mathbf{j}.$$

Thus,

$$\mathbf{v}(t) = \frac{t^2}{2}\mathbf{i} + (e^t - 1)\mathbf{j} + (2 - e^{-t})\mathbf{k}.$$

Now integrate  $\mathbf{v}(t)$  to get  $\mathbf{r}(t)$ :

$$\begin{aligned}\mathbf{r}(t) &= \int \mathbf{v}(t) dt \\ &= \int \left( \frac{t^2}{2}\mathbf{i} + (e^t - 1)\mathbf{j} + (2 - e^{-t})\mathbf{k} \right) dt \\ &= \frac{t^3}{6}\mathbf{i} + (e^t - t)\mathbf{j} + (2t + e^{-t})\mathbf{k} + \mathbf{C}_2.\end{aligned}$$

Use  $\mathbf{r}(0) = \mathbf{j} + \mathbf{k}$ :

$$\mathbf{r}(0) = \langle 0, 1, 1 \rangle + \mathbf{C}_2 = \langle 0, 1, 1 \rangle \Rightarrow \mathbf{C}_2 = \mathbf{0}.$$

Therefore,

$$\mathbf{r}(t) = \frac{t^3}{6}\mathbf{i} + (e^t - t)\mathbf{j} + (2t + e^{-t})\mathbf{k}.$$

10. At a point  $P$ , the velocity and acceleration vectors of a particle moving in space are  $\mathbf{v} = \mathbf{i} - \mathbf{j} + \mathbf{k}$  and  $\mathbf{a} = 2\mathbf{i} - 3\mathbf{k}$ , respectively. Determine the curvature of the particle's path at  $P$ .

To find the curvature  $\kappa$ , we use the formula

$$\kappa = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3}.$$

First compute the cross product:

$$\mathbf{v} \times \mathbf{a} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -1 & 1 \\ 2 & 0 & -3 \end{vmatrix} = 3\mathbf{i} + 5\mathbf{j} + 2\mathbf{k}.$$

Now,

$$|\mathbf{v} \times \mathbf{a}| = \sqrt{3^2 + 5^2 + 2^2} = \sqrt{38} \quad \text{and} \quad |\mathbf{v}| = \sqrt{1^2 + (-1)^2 + 1^2} = \sqrt{3}.$$

Therefore, at point  $P$ ,

$$\kappa = \frac{\sqrt{38}}{(\sqrt{3})^3}.$$

## SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

**Arc Length.** If  $\mathbf{r}$  is a smooth vector function on the interval  $[a, b]$ , then the arc length of the curve described by  $\mathbf{r}$  between the points with position vectors  $\mathbf{r}(a)$  and  $\mathbf{r}(b)$  is

$$L = \int_a^b |\mathbf{r}'(t)| dt.$$

**Unit Tangent, Unit Normal, and Binormal Vectors.** Let  $\mathbf{r}$  be a smooth vector function. The **unit tangent vector** at time  $t$  is defined by

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}.$$

The **(principal) unit normal vector** is defined by

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{|\mathbf{T}'(t)|}.$$

The **binormal vector** is defined by

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t).$$

The vectors  $\mathbf{T}$ ,  $\mathbf{N}$ , and  $\mathbf{B}$  are orthogonal unit vectors.

**Curvature.** The **curvature** of a smooth curve  $\mathbf{r}(t)$  is given by

$$\kappa(t) = \left| \frac{d\mathbf{T}}{ds} \right| = \frac{|\mathbf{T}'(t)|}{|\mathbf{r}'(t)|} = \frac{|\mathbf{r}'(t) \times \mathbf{r}''(t)|}{|\mathbf{r}'(t)|^3}.$$

Intuitively, curvature is a scalar quantity that measures how sharply a curve bends.

**Position, Velocity, and Acceleration.** If a particle moves in space so that its position vector at time  $t$  is  $\mathbf{r}(t)$ , then the **velocity vector** of the particle is  $\mathbf{v}(t) = \mathbf{r}'(t)$ , its **speed** is  $|\mathbf{v}(t)|$ , and its **acceleration vector** is  $\mathbf{a}(t) = \mathbf{r}''(t)$ .

In terms of velocity and acceleration, the third curvature formula above may be written as

$$\kappa(t) = \frac{|\mathbf{v}(t) \times \mathbf{a}(t)|}{|\mathbf{v}(t)|^3}.$$

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### Suggested Textbook Problems

Section 13.2: 1-44, 49-52

Section 13.3: 1-30, 40-45, 51, 52

Section 13.4: 1-32, 36